Supporting Information

Dynamic Vibration-coupled Energy Transfer for Boosting ORR Catalyst in Fuel Cell

*Chenjia Liang†, Jun Yao†, Xiaoxia Hou†, Liang Li, NingZe Gao, Haoyu Lu, Ruiyao Zhao, Xiangke Guo, Nianhua Xue, Luming Peng, Xuefeng Guo, Yan Zhu, Weiping Ding**

Key Lab of Mesoscopic Chemistry, School of Chemistry and Chemical Engineering, Nanjing University, Nanjing 210023, China

* Corresponding authors. E-mail address: Dingwp@nju.edu.cn

† Equal contribution author

Experimental Section

Chemicals

All chemicals were purchased from Aldrich and TCI Shanghai as well as used without further purification (Guaranteed reagent). Comercial PtCo/C (20 wt.%, atom rate = 1:1) and Pt/C (40 wt.%) catalyst were purchased from Premetek and Johnson Matthey as benchmark. High-purity N_2, O_2, H_2, 5% H_2/N_2, were purchased from Nanjing Tianze Gas CO, LTD.

Preparation of tert-butylsulfonylcalix[4]arene

The synthesis of tert-butylthiacalix[4]arene was achieved by dissolving 2.15 mol tert-butylphenol, 4.312 mol sulfur (S_8), and 1.107 mol NaOH in 40 mL dimethoxyethane by ultrasonic. Under nitrogen protection, the reaction solution was then slowly heated to 230 ^oC and maintained at this temperature for 3 h. After cooling the reaction to approximately 60 ^oC, 70 mL anhydrous ether, 18 mL toluene, and 70 mL 4 M sulfuric acid were added. The solution was stirred for 2 h, followed by filtration and recrystallization to yield a white solid. In the next step, tert-butylsulfonylcalix[4]arene was obtained by oxidizing 1.38 mmol tert-butylthiacalix[4]arene with 20 mmol $NaBO_3 \cdot 4H_2O$ in 30 mL of chloroform and 50 mL of acetic acid at 50 ^oC for 24 h. After the reaction, 50 mL of saturated NaCl and 20 mL of chloroform were added for extraction. The organic phase was subjected to rotary evaporator to afford the tert-butylsulfonylcalix[4]arene as a white powder.

Preparation of tBuC[4]A-PtCo/C catalyst

50 mg of PtCo/C (Premetek) was uniformly dispersed in 8 mL DMF, and then 2 mL DMF and triethylamine solution containing 100 mg/mL tert-butylsulfonylcalix[4]arene was added. The mixture was stirred for 12 hours, filtered, and washed with DMF and methanol, followed by drying to obtain the tBuC[4]A-PtCo/C catalyst. As comparison samples, tert-butylphenol was modified on the surface of PtCo/C using a similar method, replacing tert-butylsulfonylcalix[4]arene with tert-butylphenol. The use of commercial PtCo/C as the starting material exclude batch to batch errors in small amounts of synthesis in the laboratory.

Physical Characterization

DRIFT spectra were collected on a Thermo Fisher Nicolet iS5 spectrometer equipped with a diffuse reflectance accessory. Powder samples were loaded directly into the sample cup. Spectra

were recorded at room temperature in air with the following parameters: resolution = 4 cm^{-1}, number of scans = 32, mirror velocity = 0.4747, aperture = 80.00, and gain = 1.0. Background correction was performed using the OMNIC software.

XPS measurements were performed on a Thermo Scientific K-Alpha spectrometer using a monochromatic Al Kα source (1486.6 eV). The analysis chamber pressure was below 5.0×10^{-7} mBar, and charge compensation was achieved using a dual-beam (electrons/Ar^+ ions) flood gun. Survey scans used a pass energy of 50 eV and step size of 1.0 eV; high-resolution scans used a pass energy of 20 eV and step size of 0.1 eV. All binding energies were referenced to the C 1s peak at 284.8 eV. Data fitting employed a Shirley background and Voigt (mixed Gaussian-Lorentzian) peak profiles.

HRTEM images and EDS analyses were obtained using an FEI Talos microscope operated at 200 kV (Ceta CCD camera, electron dose ~5 nA). Samples were prepared by dispersing the catalyst powder in ethanol via ultrasonication for 15 min, followed by drop-casting the suspension onto a copper grid and drying in air. Post-ADT samples for TEM analysis were recovered by carefully scraping the catalyst layer from the electrode and re-dispersing it in ethanol using the same procedure.

NMR spectra were recorded on a Bruker 600 MHz spectrometer. Samples were dissolved in $CDCl_3$. Spectra were acquired with 16 scans, a relaxation delay of 1 s, and at 298 K. For the post-stability analysis, tBuC[4]A was recovered from the test solution by extraction with dichloromethane, drying over anhydrous Na_2SO_4, and recrystallization from toluene prior to measurement.

Co K-edge and Pt L_3-edge X-ray absorption spectra were collected at the BL14W1 beamline of the Shanghai Synchrotron Radiation Facility (SSRF) in transmission mode using a Si(111) double-crystal monochromator. Homogeneous powder samples were pressed into thin pellets. Data processing (energy calibration, background subtraction, normalization, EXAFS extraction) was performed using the Demeter software package. EXAFS oscillations were k^3-weighted and Fourier-transformed. Detailed fitting parameters are provided in Tables S1 and S2. Wavelet transform analysis was conducted using the HAMA Fortran code with a Morlet mother wavelet ($\sigma = 1$, $\kappa = 10$).

Inductively coupled plasma emission spectrometry (ICP) experiments were performed with an Agilent 5110 (OES) system. Approximately 20 mg of catalyst was digested in a mixture of HNO_3/HCl using a microwave digestion system. The resulting solution was diluted 10-fold with deionized water prior to analysis. Calibration was performed using certified single-element standard solutions.

In-situ Raman spectroscopy was tested using LabRAM HR Evolution in O_2-saturated 0.1 M $HClO_4$ electrolyte with a stationary liquid-solid interface on the working electrode. Measurements were performed with a 532 nm laser (~2.5 mW at sample), a 50× objective, and a 600 grooves/mm grating. Spectra were collected with an acquisition time of 5 s and 5 accumulations per point.

Electrochemical Characterization

The electrochemical tests were performed using a three-electrode system on an electrochemical workstation (CHI 760e). The working electrode was prepared by dispersing 5 mg catalyst powder in a mixture of isopropanol and 5 wt% Nafion (v / v = 1.97 / 0.03) under ultrasonic agitation for 40 min in ice-water bath. Subsequently, the ink was transferred onto a glassy carbon electrode (φ = 5.6 mm, Pt loading ≈ 12 μg_{Pt} cm^{-2}) and dried at room temperature. Graphite electrode and $Hg/HgCl_2$ electrode were used as the counter and reference electrode in 0.1 M $HClO_4$ solution, respectively (The calibration of $Hg/HgCl_2$ electrode was carried out with Pt foil as working electrode in H_2-saturated electrolyte). Linear sweep voltammetry (LSV) was conducted at the potential of 0.6 - 1.0 V (vs. RHE) with a scan rate of 20 mV s^{-1} and *iR*-95% correction, in O_2-saturated 0.1 M $HClO_4$ electrolyte.

The electrochemical active surface area (ECSA) was obtained by integrating the hydrogen underpotential deposition peak at the potential of 0.05 - 0.35 V (vs. RHE) in the N_2-saturated 0.1 M $HClO_4$ electrolyte, assuming 0.21 C cm_{Pt}^{-2}.

The kinetic current density (J_k) for ORR was calculated following:

$$J_k=\frac{J_d \cdot J_0}{J_d \times J_0} \qquad \text{Equation (1)}$$

Where J_d is limiting diffusion current density and J_0 is apparent current density. Mass activity and Tafel slope both were calculated by J_k, and specific activity was calculated by ECSA.

The TOF of cathode process on PtCo NPs was calculated following:

$$TOF=\frac{J_k}{nN_sF} \qquad \text{Equation (2)}$$

Where n is the electron transfer number, F is Faraday constant (96485 C mol^{-1}), N_s is the number of active surface atoms, measured by low-temperature H_2-O_2 titration according to Figure S4, *TOF* is turnover frequency, the number of electrons converted per second over a surface atom.

In-situ ATR-SEIRAS measurements

The *in-situ* ATR-IR measurements were conducted using a PerkinElmer Spectrum (*Hefei in situ technology Co., Ltd. China*) in O_2-saturated 0.1 M $HClO_4$ electrolyte without any pretreatment, from OCP to 0.5 V (vs. RHE), at a spectral resolution of 4 cm^{-1}. The Pt loading on the electrode is approximately consistent with the three-electrode system test. The reference spectrum is measured at open circuit potential (~ 0.6 V vs RHE) in N_2-saturated 0.1 M $HClO_4$. All spectra were converted into absorbance defined as -log (I/I_0), in which I and I_0 represent sample spectra and reference spectrum, respectively.

PEMFCs Test

Membrane electrode assembly was prepared with carbon paper and proton exchange membrane (Gore M765.08), without any treatment, coated by a catalyst, through hot pressing at 140 ^oC for 2 min. Among them, 'catalyst ink' was composed of catalyst powder, isopropanol, and Nafion solutions, as well as are sonicated until no obvious powder deposits are observed at the bottom of the cup before spraying. 'Catalyst ink' was sprayed onto proton exchange membrane through ultrasonic atomization and compressed air gas guidance. Commercial PtCo/C or tBu[4]-PtCo/C was used as cathodic catalysts (Pt loading ≈ 0.1 mg cm^{-2}) as well as commercial Pt/C (40 wt.%) on anode. The cathode is fed with high-purity oxygen with 100% R.H., 150 kPa, and the flow rate is controlled by a mass flow meter (AST10-DL) at 500 sccm. Polarization curve and 30k-cycle ADT (3 s for 0.95 V and 3 s for 0.6 V) were measured using an electrochemical workstation (CS2020B) and current amplifier. Except for the calculation of mass activity, there is no IR correction for the output power and activation energy of PEMFC.

Theoretical calculation

DFT calculations: DFT calculations are performed via the Vienna ab initio simulation package (VASP)[1,2]. Ion-electron interactions are described by projection-augmented waves (PAWs). The exchange-related functional is represented by the Perdew-Burke-Ernzerhof (PBE) functional under the Generalized Gradient Approximation (GGA)[3]. The kinetic energy cut-off for

geometrically optimized was set to 500 eV, and a (3 × 3 × 1) k-point was used for the sampling of the Brillouin zone. The structure was optimized until the energy and force reached the convergence thresholds of 10-5 eV and 0.03 eV/Å, respectively. The dispersion correction of DFT-D3 was introduced into the whole calculation process. The Pt_1Co_1 alloy was cleaved to exposed the 111 surface, and tBuC[4]A was modified at the Co site according to the synchrotron radiation results. The vacuum layer thickness was set to 20 Å to fully avoid interactions caused by periodic mirroring.

MD simulations: The entire molecular dynamics simulations were performed using the GPUMD software package[4]. Our simulation cells consist of 6893 atoms. To achieve the best balance between accuracy and cost of massively parallel computing, we used neuroevolutionary machine learning potentials formulated by the method referenced to Fan et al[5]. The training dataset was derived from *ab initio* molecular dynamics simulations and fine-tuned using parameters from the recently published General Machine Learning Potential NEP89[6].The generation was set to 50000 until the loss function converges, and the Ziegler-Biersack-Littmark(ZBL) potential, which includes an additional switching function that ramps the energy, force, and curvature smoothly to zero between an inner and outer cutoff was introduced when the distance is less than 2 Å to correct the errors in the force field when atomic distances are very close. The potential energy due to a pair of atoms at a distance r_{ij} is given by:

$$E_{ij}^{ZBL}=\frac{1}{4\pi\epsilon_0}\frac{Z_iZ_je^2}{r_{ij}}\phi(r_{ij}/a)+S(r_{ij}) \qquad \text{Equation (3)}$$

$$a=\frac{0.46850}{Z_i^{0.23}+Z_j^{0.23}} \qquad \text{Equation (4)}$$

$$\phi(x)=0.18175e^{-3.19980x}+0.50986e^{-0.94229x}+0.28022e^{-0.40290x}+0.02817e^{-0.20162x} \qquad \text{Equation (5)}$$

where e is the electron charge, Z is the atomic number. The cut off distance for angles and radial were set to 6 and 5 Å, respectively, which has proven to be a general choice[7]. The number of radial and angular descriptors was set as 5, and the spherical harmonics in the angular descriptors were expanded to fifth order. The number of neurons in the implicit neural network was set to 100 to ensure the highest calculation speed while maintaining calculation accuracy as much as possible. To allow for further fine-tuning of the potential function later, we selectively ignore the L1 regularization parameters to prevent them from disappearing during the fitting process of the potential. The weights for energy and force were set to 100%, while the weight for the virial was set to 0.1 to prevent the training potential function's overall quality from being

affected by significantly erroneous virial reference values. The batch size was 10000, which is the maximum within the available memory range.

The calculation of heat flow and surface phonon properties was carried out using the Homogeneous nonequilibrium molecular dynamics (HNEMD) method and solved by spectral decomposition[8]. In HNEMD simulation, heat flow i is defined as:

$$Q=<\sum_i W_i \cdot v_i> \qquad \text{Equation (6)}$$

$$W_i=\sum_{j\neq i} r_{ij} \otimes \frac{\partial U_j}{\partial r_{ji}} \qquad \text{Equation (7)}$$

Where, v_i represents the velocity of atom i, w_i is the virial of atom i, r_{ji} represents the atomic distance between j and i, and U is the force exerted by the atom at this time. In this definition, the time difference between the potential tensor W_i and the velocity v_i at the time of measurement can be defined as the virial-velocity cross-correlation function (VCF), which is described as:

$$K(t)=<\sum_i W_i(0) \cdot v_i(t)> \qquad \text{Equation (8)}$$

The thermal conductivity tensor can be obtained by performing the Fourier transform on the VCF to obtain the following spectral decomposition of the heat flow:

$$\kappa_{\mu\nu}=\int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} \kappa_{\mu\nu}(\omega)$$
$$\frac{2}{VT}\int_{-\infty}^{+\infty} e^{i\omega t} K^{\mu}(t)dt=\sum_{\nu} \kappa_{\mu\nu}(\omega) F_e{}^{\nu} \qquad \text{Equation (9)}$$

Where, F_e is the driving force, ν is the frequency of phonons, $k_{\mu\nu}$ represents the component of heat conduction in a specific direction, and ω represents the frequency of heat fluctuations[9]. When the size of the heat transport in a specific direction is taken into account, that is, the vector is not considered, the thermal conductivity can be spectrally resolved into：

$$G=\int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} G(\omega),$$
$$G(\omega)=\frac{2}{V\Delta T}\int_{-\infty}^{+\infty} e^{i\omega t} K(t)dt. \qquad \text{Equation (10)}$$

The phonon density of states (PDOS) is defined as[10]:

$$\rho(\omega)=\int_{-\infty}^{\infty} dt e^{i\omega t} MVAC(t) \qquad \text{Equation (11)}$$

Where $MVAC(t)$ can be expression as the mass-weighted velocity auto- correlation function (MVAC) with the following formula:

$$MVAC(t)=\frac{1}{k_B T}\sum_i m_i \langle \mathbf{v}_i(0)\cdot\mathbf{v}_i(t)\rangle \qquad \text{Equation (12)}$$

The energy utilization efficiency is defined as:

$$\textit{Energy Utilization Efficiency} = \frac{\textit{Vibrational Energy Gain of tBuC[4]A}}{\textit{Exothermic Energy of } O_2 \textit{ Adsorption}} \times 100\% \qquad \text{Equation (13)}$$

Where *"Vibrational Energy Gain"* is the increase in single-point energy of the tBuC[4]A molecule upon O_2 adsorption, representing the amount of adsorption energy stored as its vibrational excitation; "*O_2 Adsorption Energy*" is the calculated exothermic energy released when an O_2 molecule adsorbs on the PtCo surface.

Using a ballistic transport model for phonon diffusion with a transfer distance of ~1.6 nm (from O_2* adsorption site to OH* desorption site) and a phonon group velocity of ~3300 m s^{-1} in Pt_1Co_1[11], Phonon dissipation time ($\tau_{dissipation}$) at surface is ~0.48 ps.

Microkinetic simulations: The elementary reaction steps of ORR processes on tBuC[4]A-PtCo/C catalyst are as presented below:

(1) O_2 diffusion from solution environment to the electrolytic double layer:

$$O_2(aq) \underset{K_{-1}}{\overset{K_1}{\rightleftharpoons}} O_2(dl) \qquad \text{Equation (14)}$$

(2) O_2 adsorption on the active site:

$$O_2(dl) + * \underset{K_{-2}}{\overset{K_2}{\rightleftharpoons}} O_2* \qquad \text{Equation (15)}$$

(3) the first protonation of O_2* to give OOH*;

$$O_2* + H_2O + e^- \underset{K_{-3}}{\overset{K_3}{\rightleftharpoons}} OOH* + OH^- \qquad \text{Equation (16)}$$

(4) the second protonation of OOH* to give O*;

$$OOH* + H_2O + e^- \underset{K_{-4}}{\overset{K_4}{\rightleftharpoons}} *O + OH^- \qquad \text{Equation (17)}$$

(5) the third protonation of O* to give OH*;

$$O* + H_2O + e^- \underset{K_{-5}}{\overset{K_5}{\rightleftharpoons}} OH* + OH^- \qquad \text{Equation (18)}$$

(6) the fourth protonation of OH* to give H_2O*;

$$OH* + H_2O + e^- \underset{K_{-6}}{\overset{K_6}{\rightleftharpoons}} H_2O* + OH^- \qquad \text{Equation (19)}$$

(7) the desorption of H_2O* from the Pt site

$$H_2O* \underset{K_{-7}}{\overset{K_7}{\rightleftharpoons}} * + H_2O \qquad \text{Equation (20)}$$

where A_i is the pre-exponential factor and $\Delta G^{\#}(U)$ is the potential-dependent activation barrier of each elementary step and $K_i = A_i exp(-\frac{\partial G^{\#}(U)}{k_B T})$. Following the protocol established by Nørskov and co-workers[12], the pre-exponential factors for the desorption of O_2* and H_2O are set to 1×10^{13}. For Eley-Rideal-like (E-R) elementary steps involving reactant species from the solvent phase and the catalyst surface, the pre-exponential factor is reduced to 2.3×10^{9} to account for the significantly lower collision frequency. The entire analysis is conducted under the steady-state approximation, and mass-transport limitations of O_2 are neglected.

Calculation method about the distance between two Calix[4]arenes:

The specific surface area of the PtCo NPs (S_{PtCo}) was calculated following:

$$S_{PtCo} = 4\pi r^2 \times (4/3 \times \pi \rho r^3)^{-1} \quad \text{Equation (21)}$$

Where r is the particle size of PtCo NPs and ρ *(15.79 g·cm^{-3})* is the density of Pt_1Co_1.

Number of molecules of modified calix[4]arenes on 1 g PtCo NPs ($N_{calix[4]arene}$) was calculated following:

$$N_{calix[4]arene} = m_{calix[4]arene}/M_{calix[4]arene} \times N_A \quad \text{Equation (22)}$$

Where $m_{calix[4]arene}$ is the mass of calix[4]arenes, $M_{calix[4]arene}$ is relative molecular mass and N_A is Avogadro constant.

Assuming the area occupied ($A_{calix[4]arene}$) by a single calix[4]arene is 1 nm^2, the total area of modified calix[4]arenes ($A_{calix[4]arene}$) was calculated following:

$$S_{calix[4]arene} = A_{calix[4]arene} \times N_{calix[4]arene} \quad \text{Equation (23)}$$

The average distance between two calix[4]arene on the surface of the nanoparticles was calculated following:

$$D = (A_{calix[4]arene}/S_{PtCo})^{-0.5} \quad \text{Equation (24)}$$

Figures and Tables:

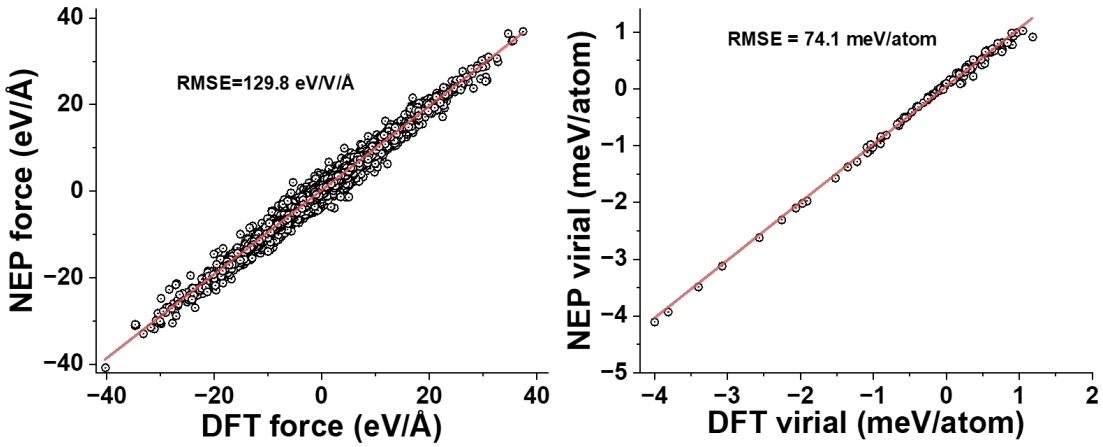


Figure S1. RMSE graph of training dataset.

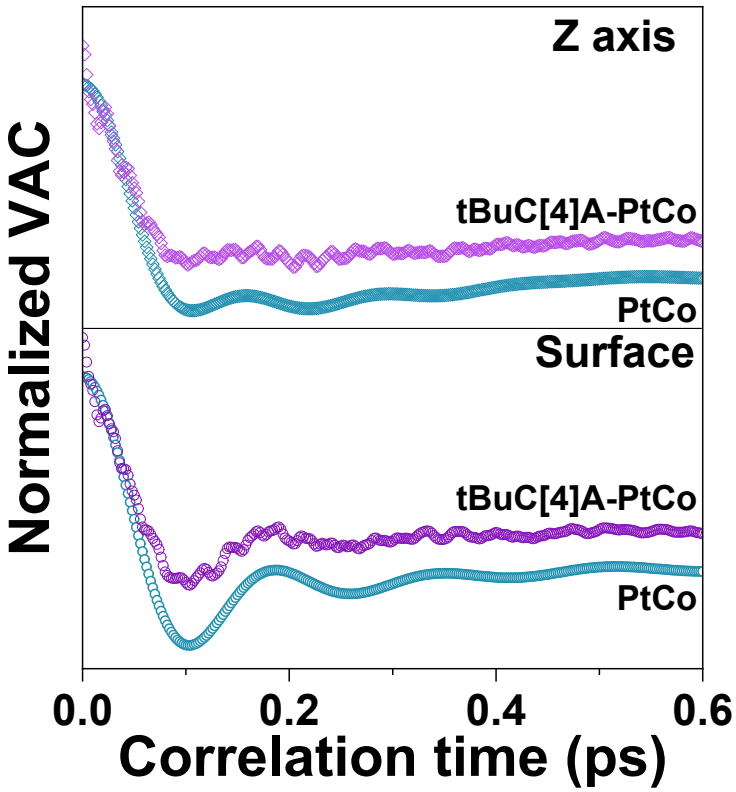


Figure S2. The velocity autocorrelation functions comparison of phonons between PtCo and tBuC[4]A-PtCo.

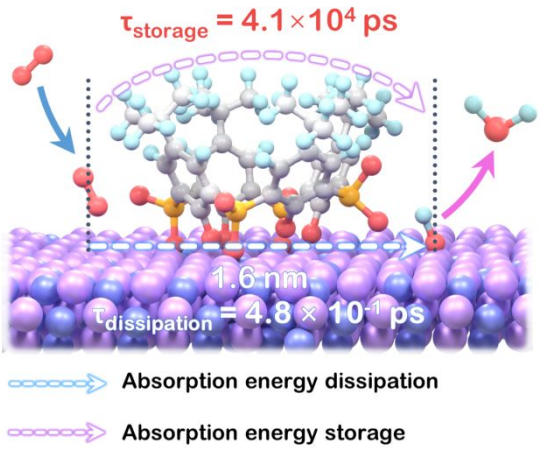


Figure S3. The comparison of absorption energy dissipation and storge.

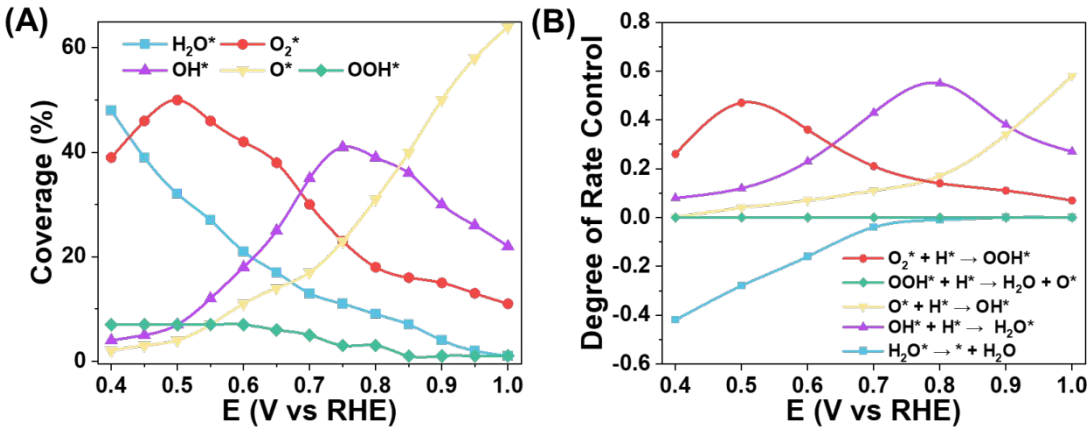


Figure S4. (A) The evolution trend of ORR intermediates coverage and (B) the degree of rate control on tBuC[4]A-Pt_1Co_1(111) by microkinetic modeling.

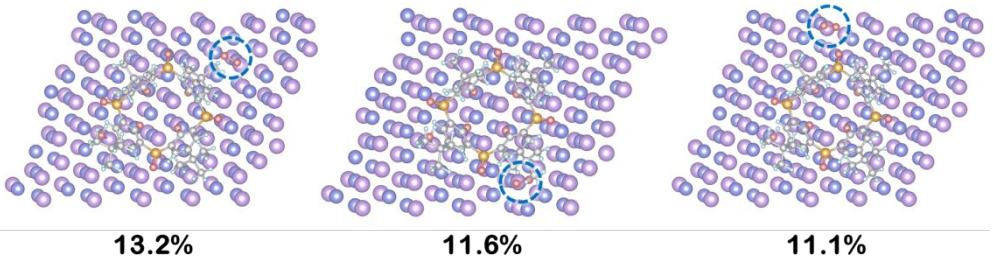


Figure S5. The energy conversion efficiency triggered by distinct O_2-adsorption sites.

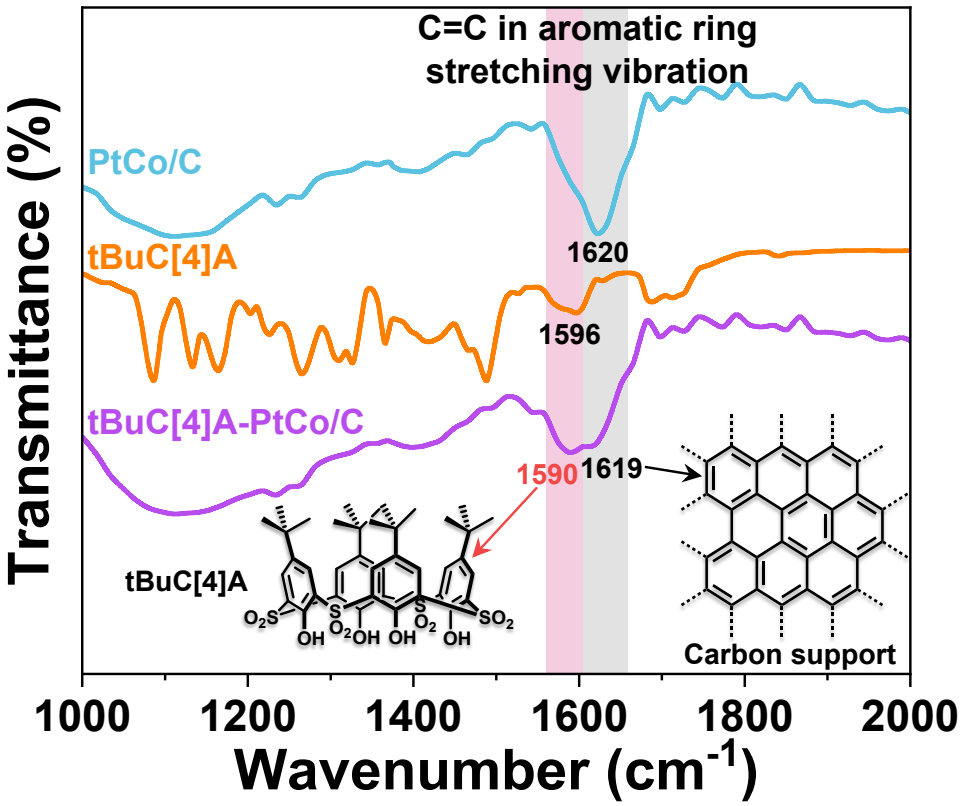


Figure S6. Diffuse reflectance infrared spectroscopy of tBuC[4]A-PtCo/C, PtCo/C and tBuC[4]A.

Note: The absorbance signals for both PtCo/C and tBuC[4]A-PtCo/C have been multiplied by a factor of 4 for clarity. The characteristic vibrational features of tBuC[4]A in tBuC[4]A-PtCo/C remain relatively weak due to the strong background absorption of the carbon black support and the low surface concentration.

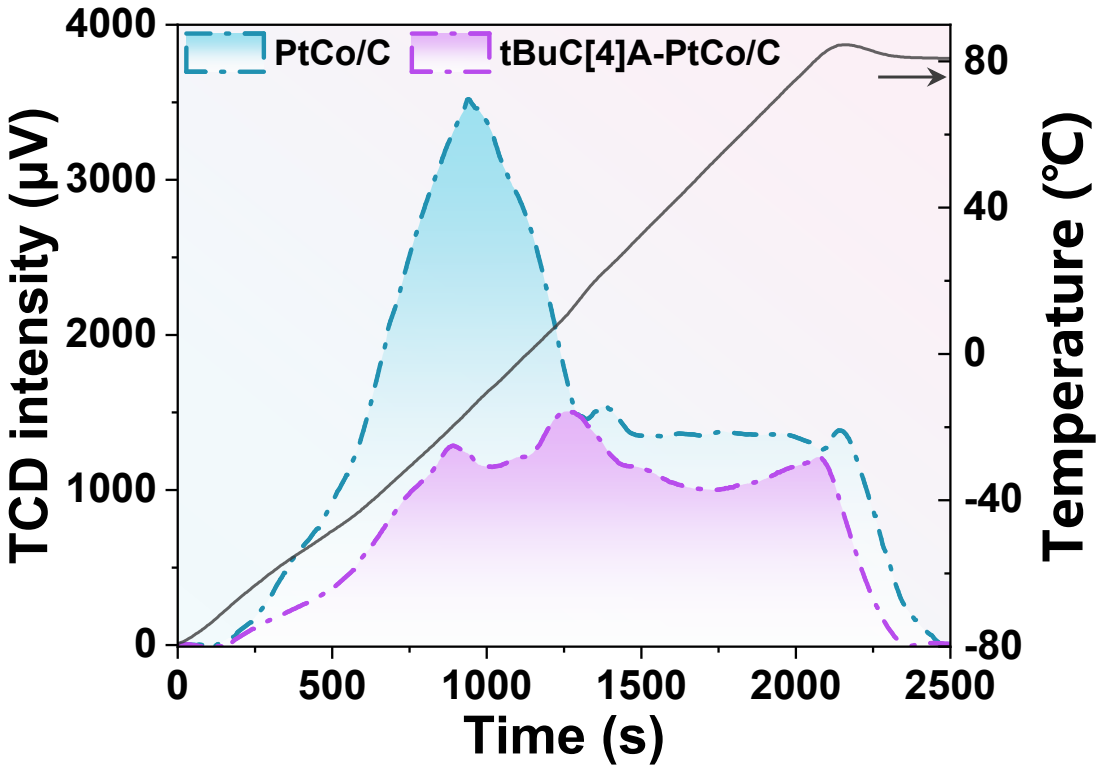


Figure S7. Low-temperature H_2-O_2 titration curve of PtCo/C and tBuC[4]A-PtCo/C.

Prior to H_2-O_2 titration measurements, the catalyst was pre-reduced in flowing H_2 at 80 °C for 1 h to remove pre-existing surface oxygen species. Subsequently, the sample was exposed to O_2 at room temperature for chemisorption, followed by purging with N_2 for 1 h to remove any physisorbed oxygen. The system was then cooled to -80 °C in an inert atmosphere, and the gas flow was switched to 5% H_2/N_2. The temperature-programmed reduction profile was recorded during heating to 80 °C while monitoring H_2 consumption. According to the consumption of H_2, the chemical adsorption content of O_2 on the surface of PtCo/C and tBuC[4]A-PtCo/C is 0.079 mmol g^{-1} and 0.04175 mmol g^{-1}, respectively

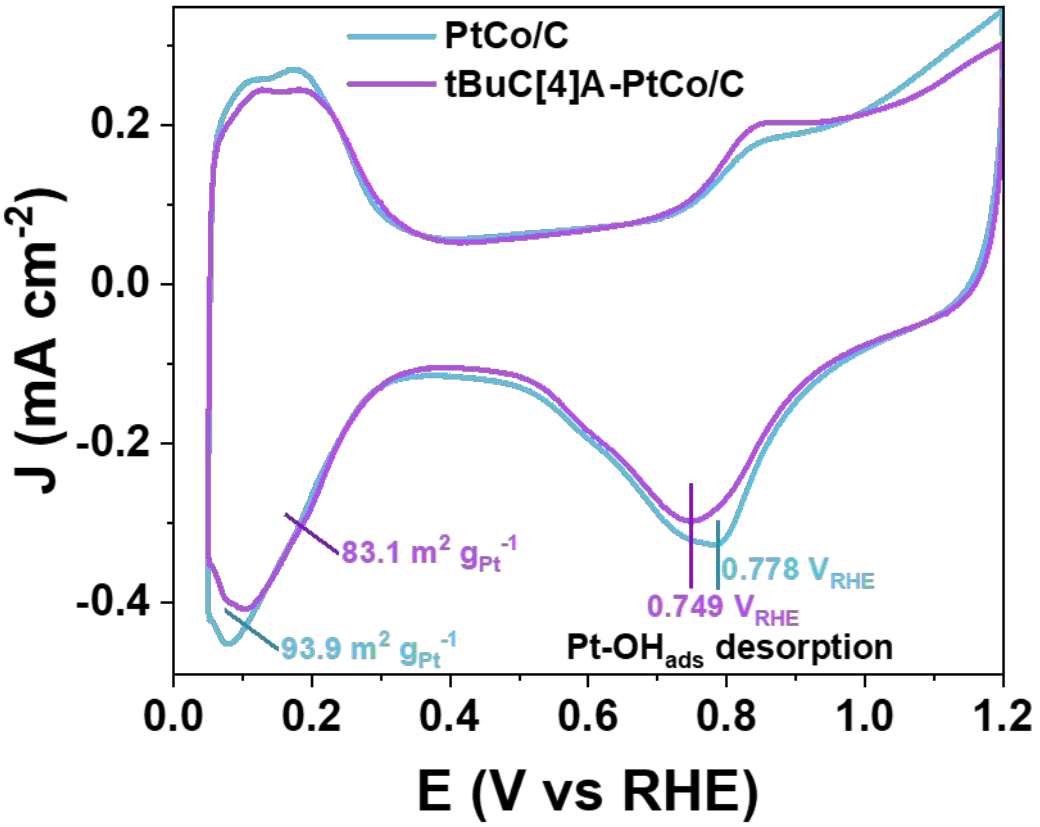


Figure S8. The CV curve of PtCo/C and tBuC[4]A-PtCo/C in N_2-saturated 0.1 M $HClO_4$ with 20 mV s^{-1}.

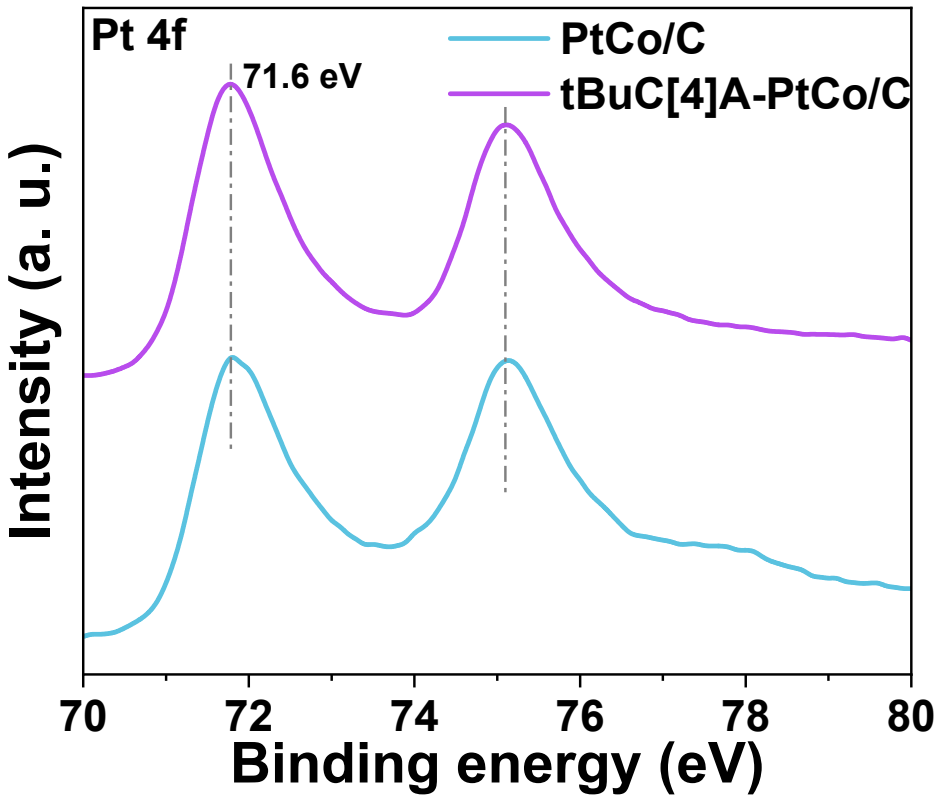


Figure S9. The XPS Pt 4f spectra of PtCo/C and tBuC[4]A-PtCo/C.

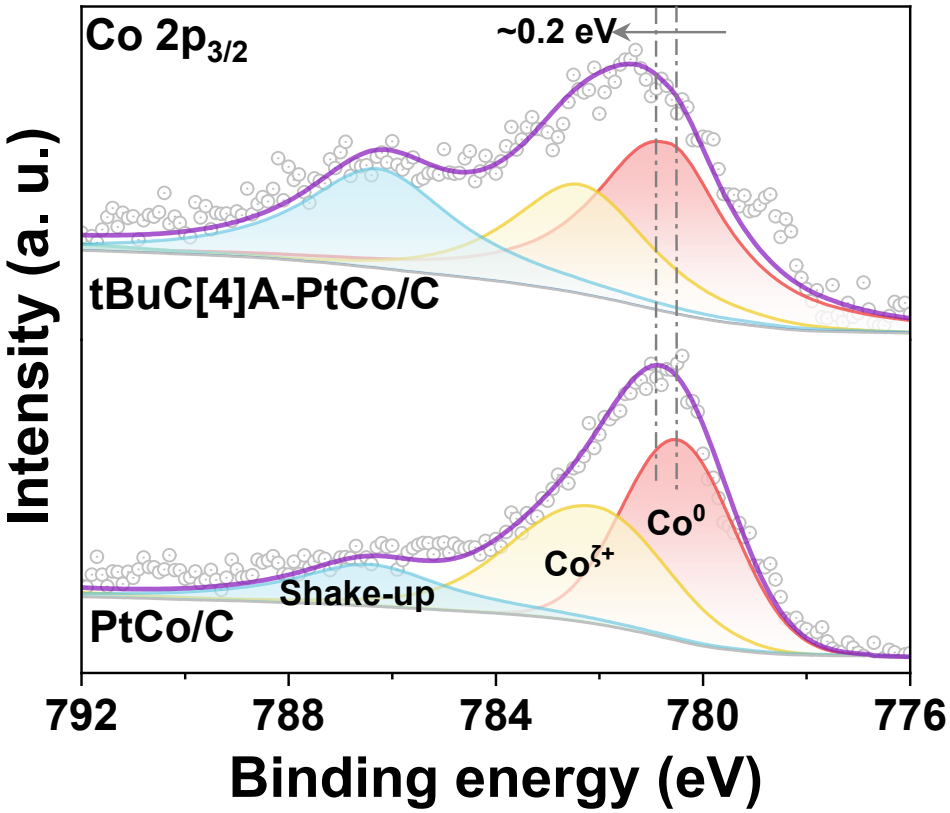


Figure S10. The XPS Co $2p_{3/2}$ spectra of PtCo/C and tBuC[4]A-PtCo/C.

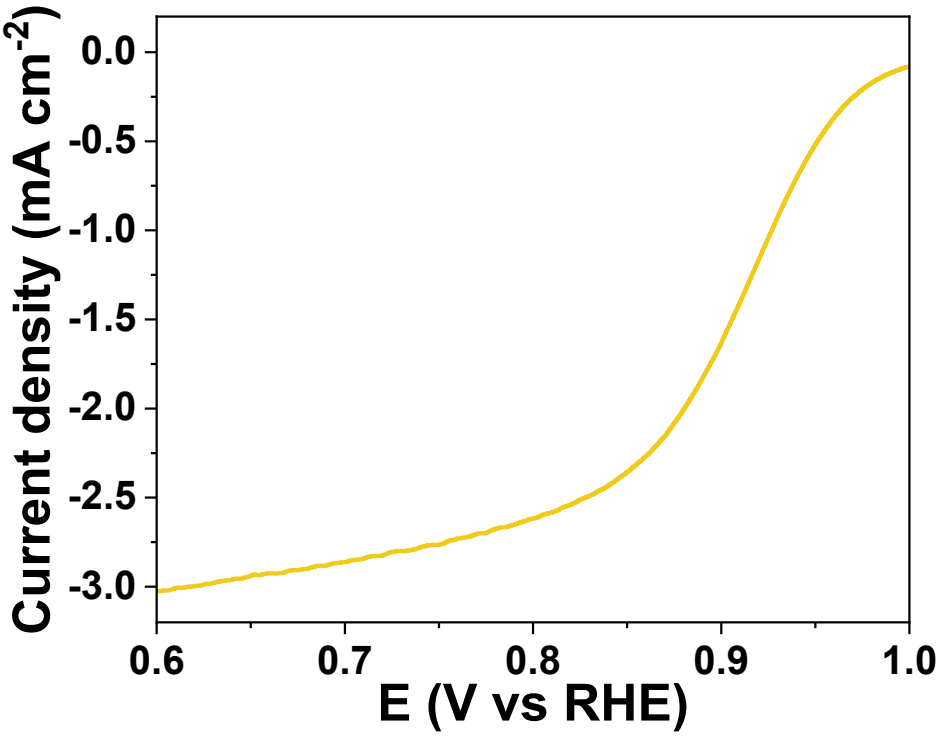


Figure S11. ORR polarization curves for tBuPh-PtCo/C in O_2-saturated 0.1 M $HClO_4$.

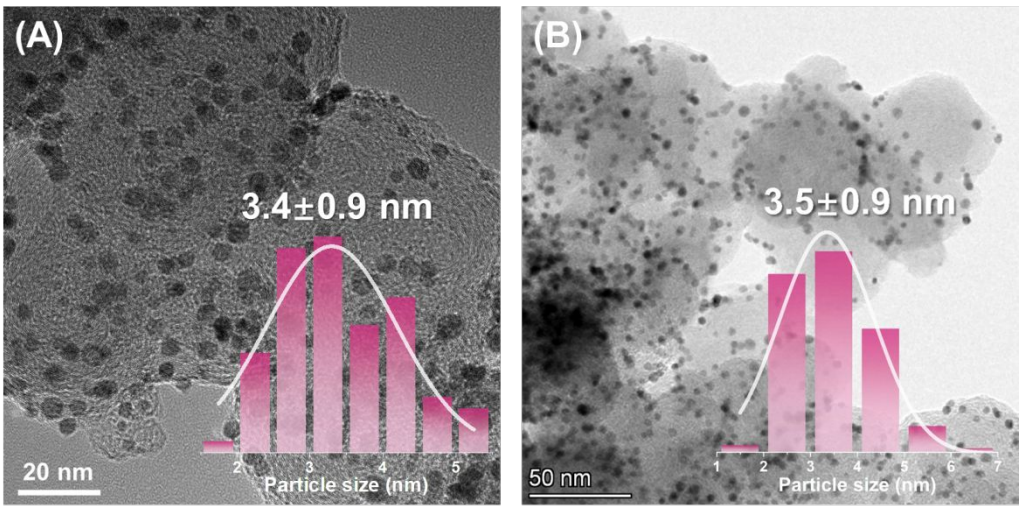


Figure S12. The HRTEM image of (A) fresh and (B) used tBu[4]-PtCo/C catalyst, undergone 30k-cycle ADT and separated from the electrode by ultrasound.

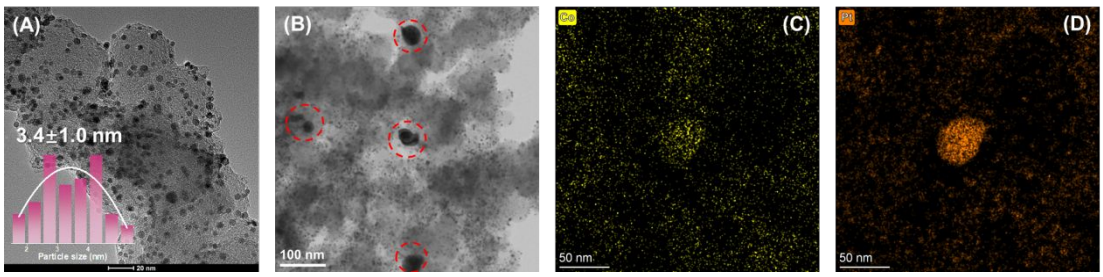


Figure S13. The HRTEM image of PtCo/C (A) before and (B) after 30k-cycle ADT; the EDX mapping of PtCo/C after 30k-cycle ADT for (C) Co and (D) Pt element.

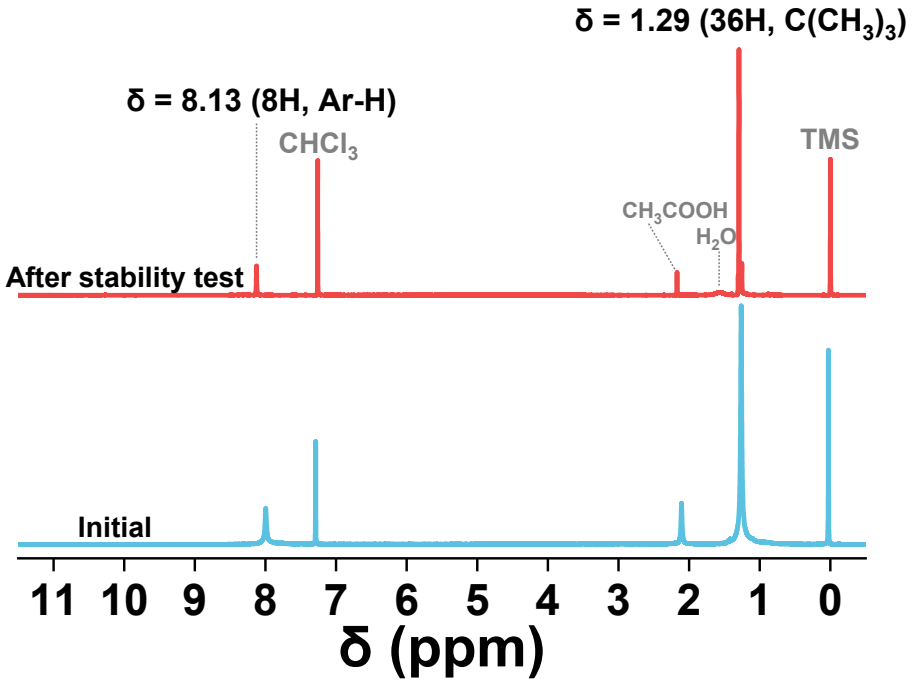


Figure S14. The 1H NMR NMR spectrum of tBuC[4]A before and after 100-hour stability tests by immersed in O_2-saturated 0.1 M $HClO_4$ at 80 °C.

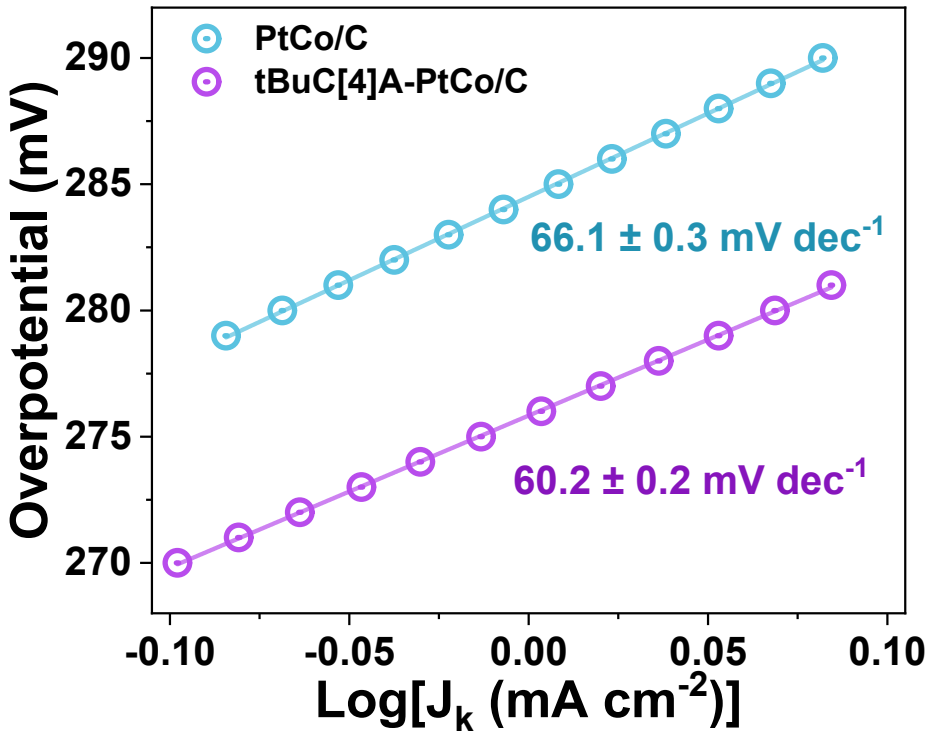


Figure S15. Tafel plots linearly fitted by *Log(J_k)* and overpotential.

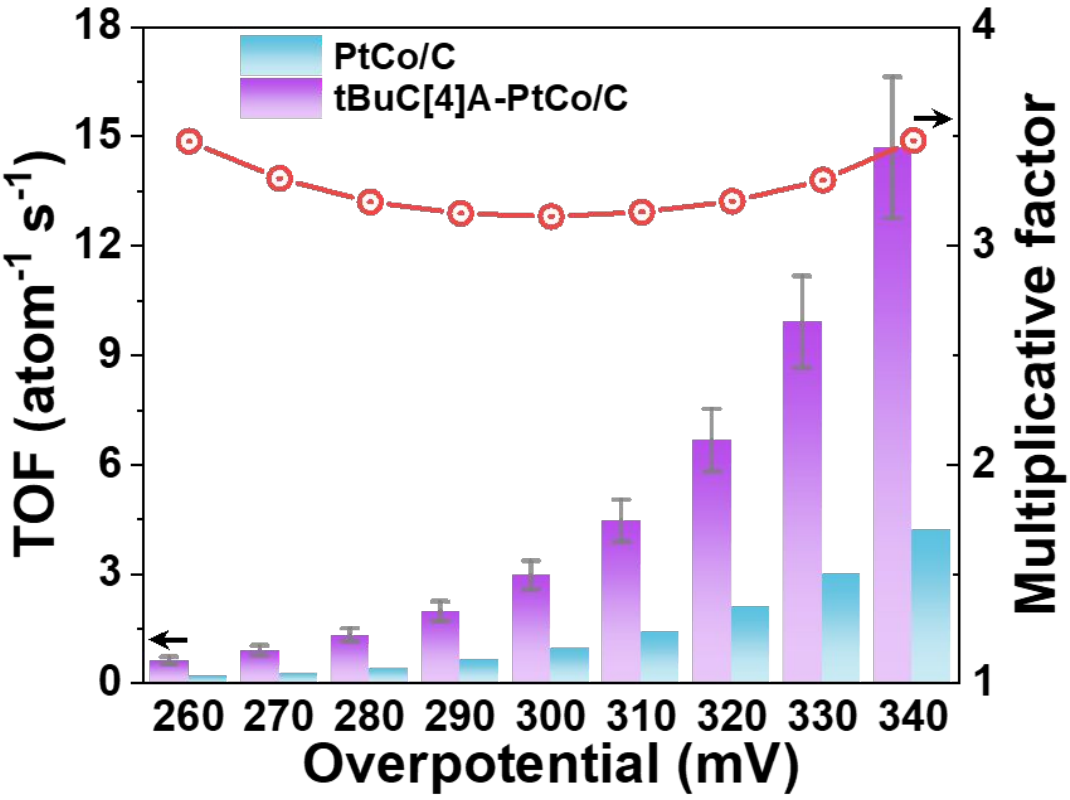


Figure S16. The TOF for PtCo and tBuC[4]A-PtCo/C, under different overpotentials, and multiplicative factors.

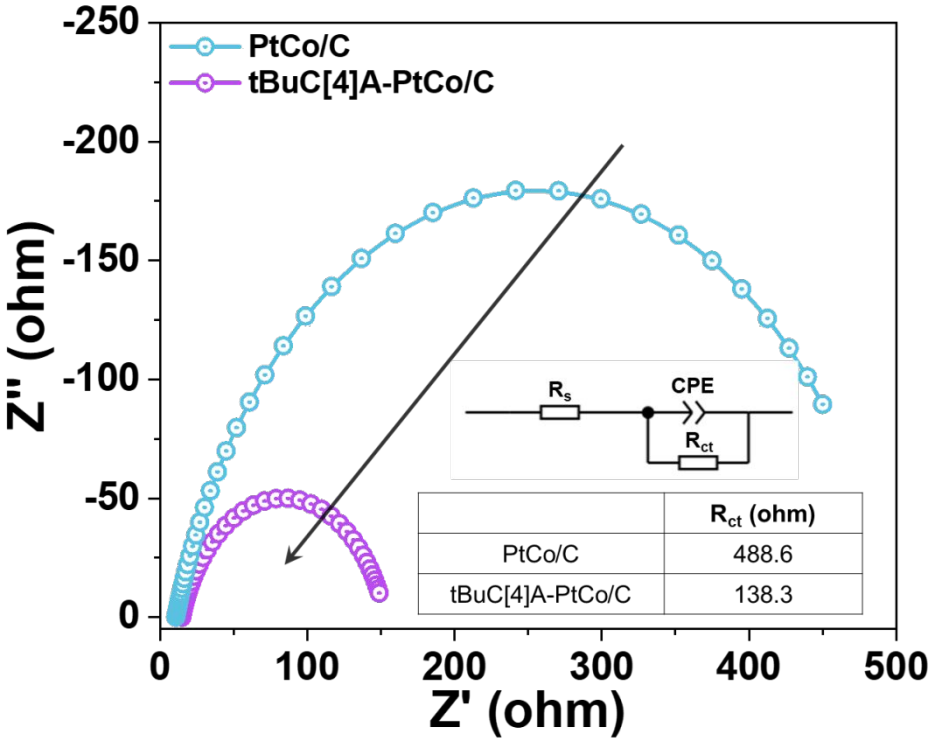


Figure S17. The electrochemical impedance spectroscopy, in O_2-saturated 0.1 M $HClO_4$ at 0.92 V_{RHE} by gas diffusion electrode, and fitted charge-transfer resistance (R_{ct}) for PtCo/C and tBuC[4]A-PtCo/C, respectively.

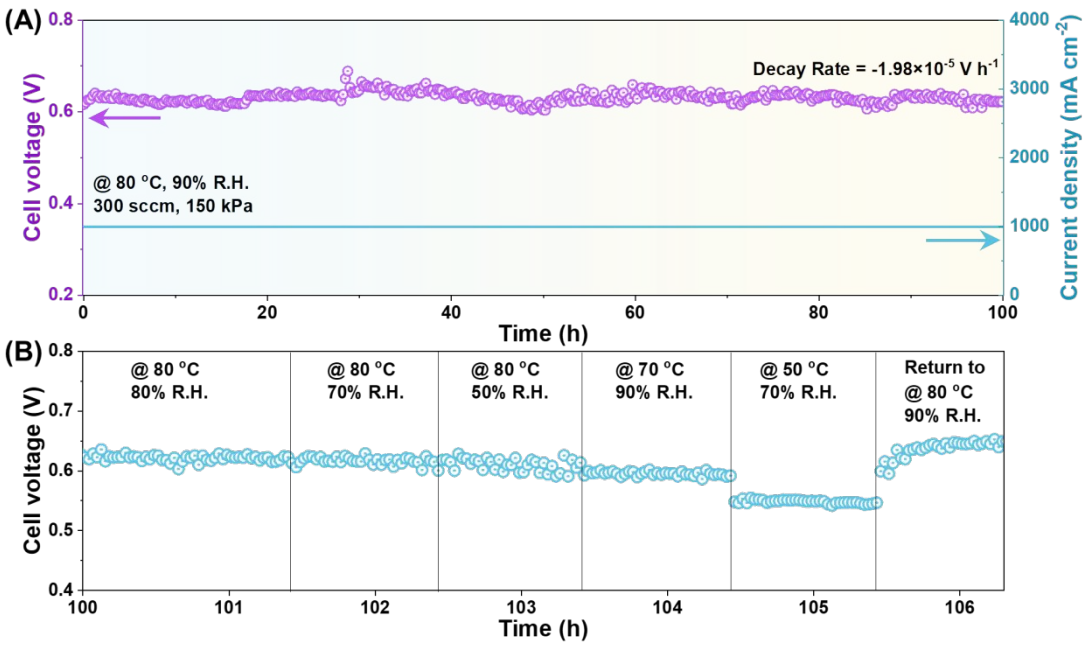


Figure S18. (A) The constant current discharge test of PEMFC with tBuC[4]A-PtCo/C as cathode. (B) The stability testing of humidity and temperature changes using the same membrane electrode assembly.

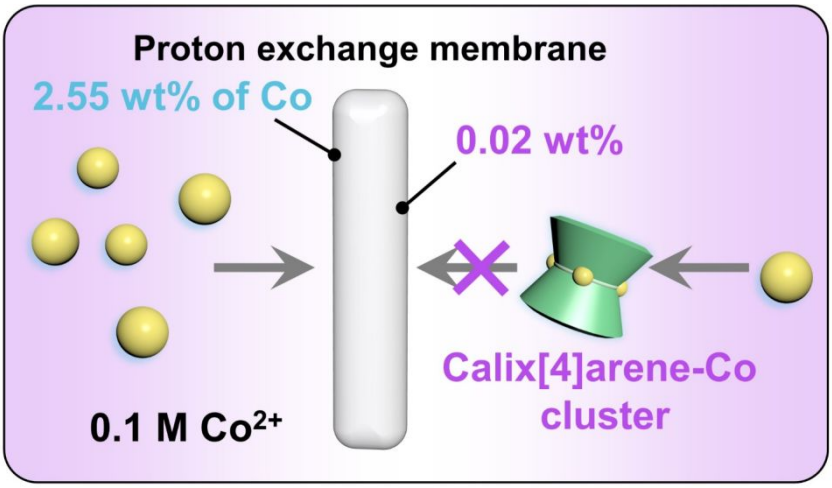


Figure S19. Schematic diagram of an extreme test to prevent leaching of metal ions into PEM.

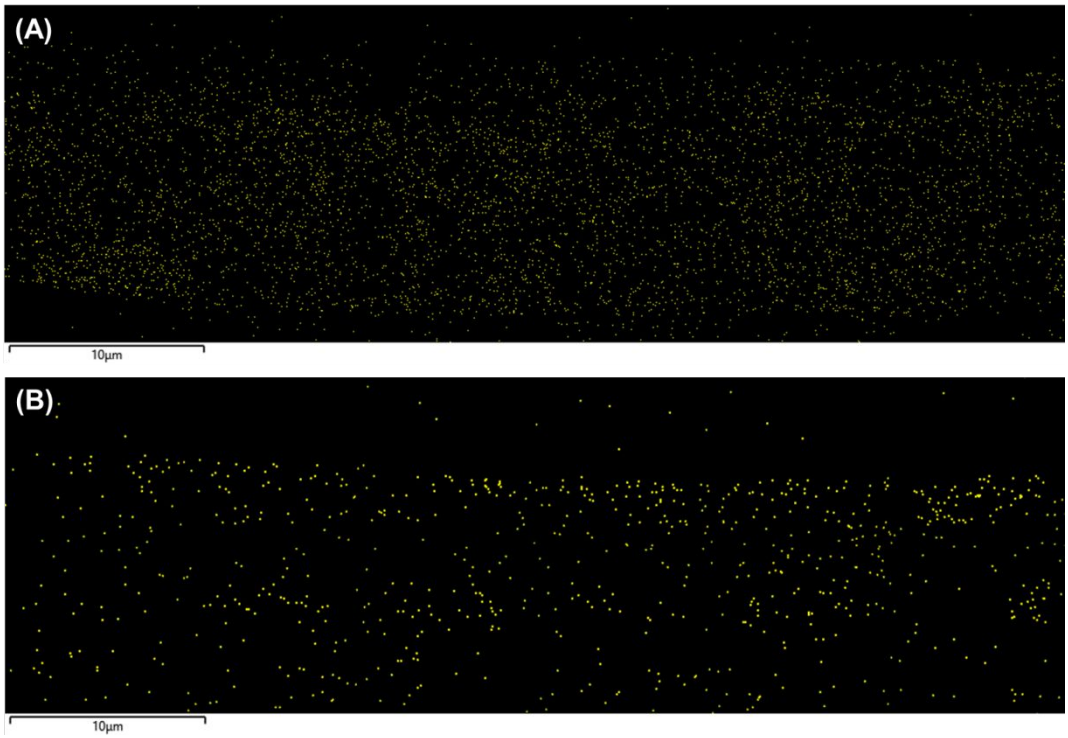


Figure S20. The EDS mapping of Co element on the PEM cross-section after extreme immersion test in (A) 0.1 M $Co(NO_3)_2$ and (B) 0.1 M $Co(NO_3)_2$ + calix[4]arene.

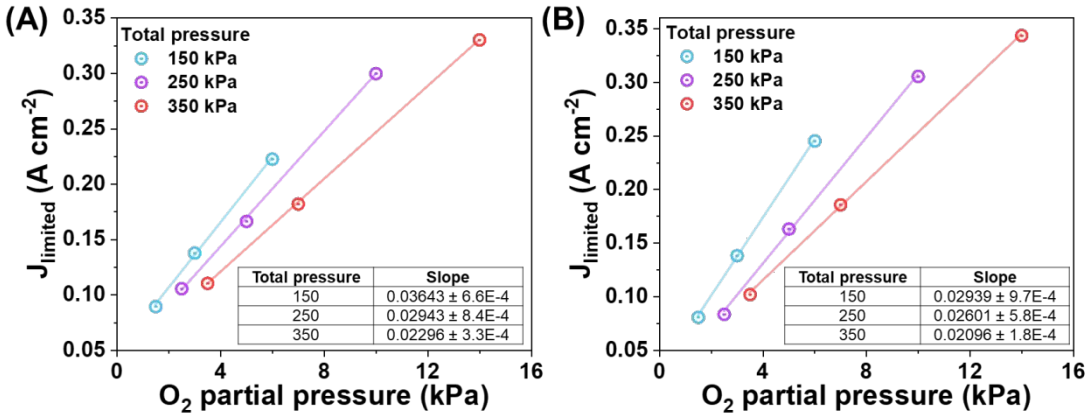


Figure S21. The relationship curve between O_2 partial pressure and the limited current density ($J_{limited}$) of PEMFC for (A) PtCo/C and (B) tBu[4]-PtCo/C.

The O_2 transport resistance (R^{Total}) and the pressure-independent parts of transport resistance ($R^{P,ind}$) in Figure 4A was calculated as following:

$$R^{Total} = 4 \times F \times P_{oxygen} \times (R \times T \times J_{limited})^{-1} \quad \text{Equation (25)}$$

$$R^{Total} = R^{P,\,dep} + R^{P,\,ind} \quad \text{Equation (26)}$$

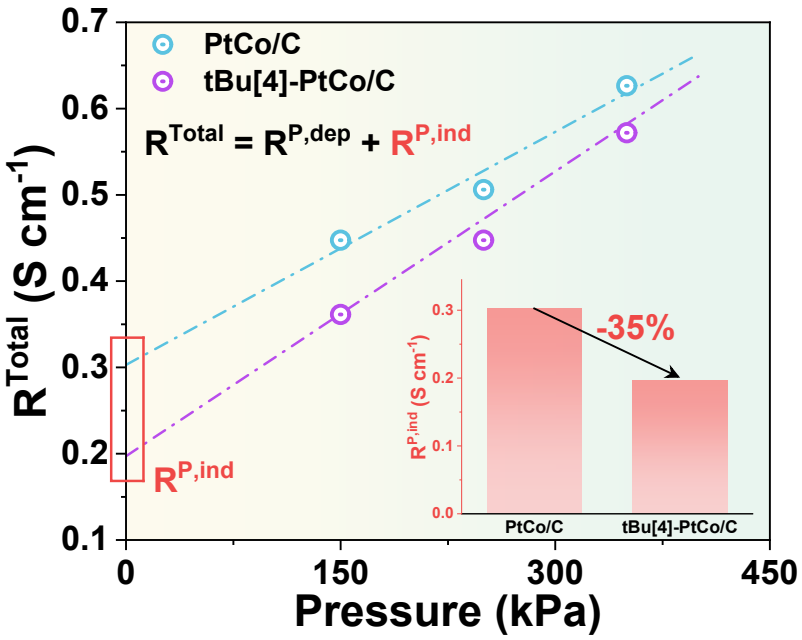


Figure S22. Comparison of the pressure-independent parts of oxygen transport resistance between PtCo/C and tBu[4]-PtCo/C

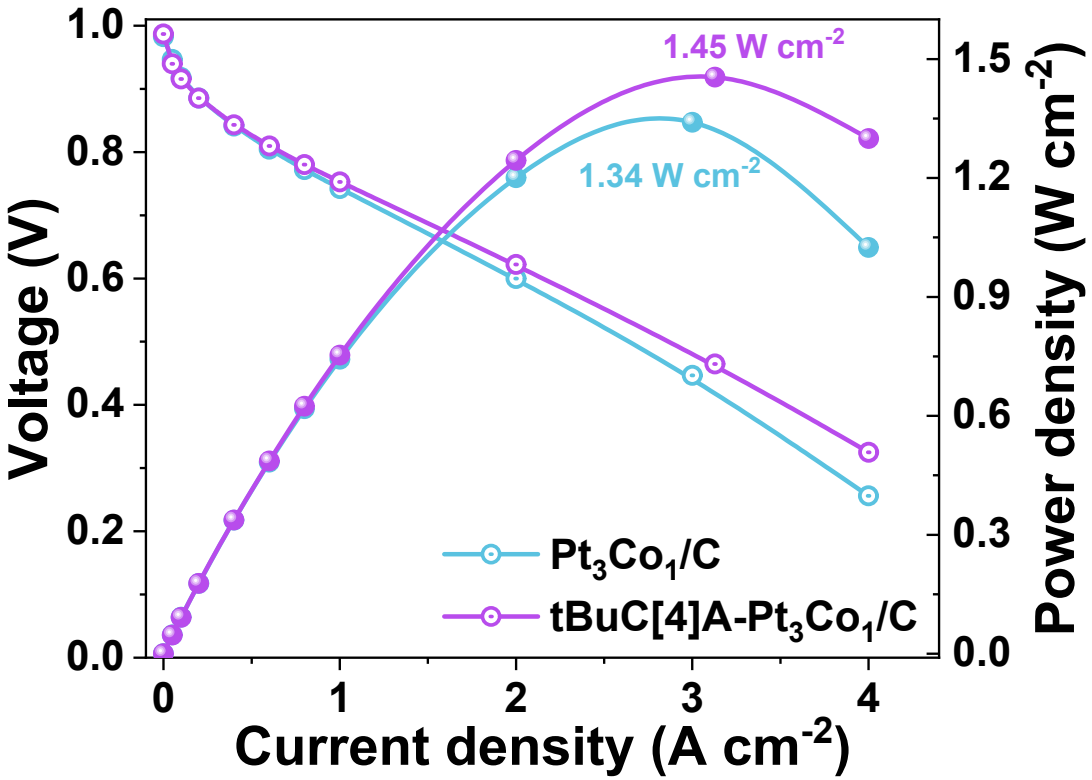


Figure S23. The PEMFC performance of Pt_3Co_1/C and tBuC[4]A-Pt_3Co_1/C. The Pt_3Co_1/C was prepared according to Huang et al[13] work.

Table S1. The EXAFS fitting parameters at the Pt L_3-edge for PtCo/C and tBuC[4]A-PtCo/C (S_0^2=0.85 from Pt-foil).

	shell	CN[a]	R[b](Å)	σ^{2c}(Å^2)	ΔE_0[d](eV)	R factor
Pt-foil	Pt-Pt	12	2.76±0.01	0.0048	7.6±0.5	0.0028
PtCo/C	Pt-O	1.1±0.3	2.07±0.02	0.0039	9.7±1.3	0.0104
	Pt-Co	4.5±0.8	2.61±0.01	0.0150		
	Pt-Pt	8.2±0.7	2.77±0.01	0.0059		
tBuC[4]A-PtCo/C	Pt-O	1.1±0.2	2.01±0.02	0.0056	8.8±1.0	0.0070
	Pt-Co	3.1±0.7	2.67±0.01	0.0139		
	Pt-Pt	7.3±0.7	2.74±0.01	0.0064		

[a]*CN*: coordination numbers; [b]*R*: bond distance; [c]σ^2: Debye-Waller factors; [d] ΔE_0: the inner potential correction. R factor: goodness of fit. Error bounds that characterize the structural parameters obtained by EXAFS spectroscopy were estimated as CN ± 20%; R ± 1%; σ^2 ± 20%.

Table S2. The EXAFS fitting parameters at the Co K-edge for PtCo/C and tBuC[4]A-PtCo/C (S_0^2=0.76 from Co-foil).

	shell	CN[a]	R[b](Å)	σ^{2c}(Å^2)	ΔE_0[d](eV)	R factor
Co-foil	Co-Co	12	2.49±0.01	0.0063	6.9±0.3	0.0008
PtCo/C	Co-O	1.7±0.2	1.97±0.01	0.0030	-3.7±1.7	0.0054
	Co-Co	3.2±0.2	2.50±0.01	0.0061		
	Co-Pt	3.1±1.0	2.67±0.01	0.0123		
tBuC[4]A-PtCo/C	Co-O	4.0±0.2	1.97±0.01	0.0052	-8.1±1.5	0.0104
	Co-Co	1.0±0.2	2.45±0.01	0.0082		
	Co-Pt	1.5±0.2	2.71±0.01	0.0020		

[a]*CN*: coordination numbers; [b]*R*: bond distance; [c]σ^2: Debye-Waller factors; [d] ΔE_0: the inner potential correction. R factor: goodness of fit. Error bounds that characterize the structural parameters obtained by EXAFS spectroscopy were estimated as CN± 20%; R ± 1%; σ^2 ± 20%.

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